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Differential Evolution-Based CNN-Transformer Fusion for Thoracic Disease Classification from Chest X-Rays

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ABSTRACT The complementary strengths of Convolutional Neural Networks (CNNs) and Transformer architectures have motivated the development of fusion strategies. However, existing fusion approaches mostly rely on heuristic methods, unified backbones, or globally shared fusion coefficients. In this study, we propose a Differential Evolution (DE)-based optimization framework for adaptive CNN–Transformer fusion in thoracic disease classification from ChestX-ray (CXR) images. The framework consists of two optimization-based fusion strategies: (i) DE-based Global Fusion (DE-GF), which optimizes a single pair of globally shared fusion weights, and (ii) DE-based Class-Specific Fusion (DE-CSF), which extends the formulation by optimizing independent fusion weights for each disease category. Extensive experiments were conducted on the NIH CXR dataset using two backbone configurations (DenseNet + ViT and DenseNet + Swin Transformer). Results demonstrate that DE-based optimization consistently improves performance over individual backbones and existing decision-level fusion methods. Although the performance gains over DE-GF are generally modest, DE-CSF achieves the highest mean AUC while providing more balanced performance across disease categories. The learned fusion coefficients vary across both disease categories and backbone combinations, suggesting that class-specific optimization can better adapt the contribution of CNN and Transformer models than globally shared fusion weights. These results demonstrate that optimization-based class-specific fusion provides an effective and flexible decision-level integration strategy for thoracic disease classification.

INDEX TERMS Chest X-ray Classification, Thoracic Diseases, CNN–Transformer ensemble, Differential Evolution, Optimization-based fusion, Class-specific weighting

I. INTRODUCTION

Chest X-ray (CXR) imaging is one of the most widely used and cost-effective modalities for the screening and diagnosis of thoracic diseases [1]–[3]. However, accurate interpretation of CXR images remains challenging due to variations in image quality, overlapping anatomical structures, and the complex visual patterns of different diseases [4]. These challenges are more evident in developing countries and rural areas, where the number of trained radiologists is insufficient to meet the increasing demand for image analysis. Artificial intelligence–based automated analysis of medical images has emerged as a promising tool to support clinical decision-making by providing rapid, consistent, and accurate assessments of CXR images [2].

Thoracic pathologies exhibit diverse heterogeneity in their visual manifestations. Some diseases, such as pulmonary nodules, masses, and atelectasis are primarily characterized by localized imaging findings, including small opacities, subtle texture variations and region-specific structural abnormalities. In contrast, other conditions, such as cardiomegaly, pleural effusion, and infiltration, are better described by global anatomical changes, including organ enlargement, fluid distribution, and diffuse spread across the lung fields. Certain diseases, such as pneumonia and pneumothorax, may present both localized abnormalities and globally distributed patterns. Convolutional Neural Networks (CNNs)-based models are effective at capturing local visual features, such as textures, edges, and small structural changes, mak-

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ing them suitable for diseases with localized manifestations. However, their ability to model global relationships across the lungs is limited. In contrast, Transformer-based models use self-attention mechanisms to capture global context, but they are less effective in representing fine-grained local details [5].

The complementary strengths of CNN and Transformer architectures have motivated extensive efforts to integrate their predictions in medical image analysis [6]. Existing approaches broadly follow two directions: unified backbone designs, such as the Swin Transformer [7], which capture both local and global dependencies within a single architecture, and integration-based strategies that combine CNN and Transformer models either at the feature stage or at the prediction stage [8], [9], [12], [13]. In this study, we use the Differential Evolution (DE) algorithm [38] as an optimization framework to fuse Transformer- and CNN-based architectures. DE provides data-driven optimization of fusion coefficients by iteratively searching for weight configurations that maximize predictive performance. We investigate two ensemble formulations: (i) Differential Evolution-Based Global Fusion (DE-GF), a conventional ensemble setting in which a single pair of globally shared fusion weights is learned to combine CNN and Transformer predictions across all disease categories; and (ii) Differential Evolution-Based Class-Specific Fusion (DE-CSF), in which independent fusion weights are assigned to each disease class.

The main contributions of this work are as follows: (i) We investigate the performance of a DE-based ensemble framework for decision-level fusion of CNN- and Transformer-based predictions and extend it to a class-specific formulation in which independent fusion weights are optimized for each disease category. The proposed framework is described in detail, including the DE-based optimization procedure and computational complexity analysis. (ii) We systematically evaluate multiple CNN–Transformer backbone combinations, namely DenseNet, MobileNet, ViT, and Swin Transformer, to assess the effectiveness of the proposed fusion framework across different architectural configurations. (iii) We compare the DE-based fusion frameworks with several decision-level fusion methods, including Ensemble-Max, Ensemble-MLP, two-layer Ensemble-MLP, Ensemble-Soft Gating, and DE-GF.

II. RELATED WORK

Deep learning–based CXR analysis has mostly relied on CNNs, such as ResNet, DenseNet, and similar variants, which have demonstrated strong performance in thoracic disease classification tasks. More recently, Vision Transformers (ViTs) have been introduced and use self-attention mechanisms to model long-range contextual relationships across image patches. A recent systematic review analyzing thirty-six medical imaging studies reported that Transformer-based approaches outperformed CNNs in more than half of the evaluated tasks and achieved comparable performance in approximately one-third of them [14]. These findings suggest that Transformer architectures have strong representational capacity for medical image analysis. However, the review also highlights that CNN-based models remain competitive and, in certain scenarios, more practical alternatives. In particular, factors such as task characteristics, dataset size and quality, and computational complexity may favor CNN-based approaches.

Given the complementary strengths of CNNs and Transformers, several studies have explored integration strategies. Unified architectures such as the Swin Transformer, aim to model both local and global dependencies within a single backbone. When evaluated on the NIH CXR dataset, Swin Transformer–based models demonstrated improved performance in certain disease categories, including Emphysema, Atelectasis, Mass, and Pneumonia, compared to earlier unimodal CNN and Transformer architectures [12]. However, these improvements were not uniformly observed across all classes, indicating that enhanced architectural design alone does not guarantee consistent gains for heterogeneous thoracic pathologies. Moreover, although Swin Transformers enhance locality modeling, they still lack a mechanism to adjust the relative importance of local and global features on a per-class basis. Similarly, hybrid CNN–Transformer frameworks integrate convolutional feature extraction with global self-attention in a unified model [8]. Although hybrid CNN–Transformer architectures are designed to use complementary local and global representations, empirical results indicate that they do not consistently outperform strong standalone backbones. In several studies, individual CNN or Transformer models achieved comparable or higher AUC values than hybrid architectures, suggesting that effective integration of CNN- and Transformer-based models remains a challenging problem.

Beyond unified backbones, several studies have explored dual-branch architectures and feature-level integration strategies to combine CNN and Transformer architectures. For instance, Wu et al. [17] proposed CTransCNN, which processes CXR images through separate CNN and Transformer branches and fuses their embedded features via an interaction module. Experimental results demonstrated that this fusion strategy improved classification performance compared to individual backbone models. Similarly, Chowdary et al. [18] proposed a dual-branch architecture that processes both the full CXR image and lung-focused regions obtained using segmentation masks. The representations from the two branches are fused at the feature level to produce the final classification. At the decision level, Li et al. [13] proposed a late-fusion ensemble framework that combines DenseNet121 and Swin Transformer predictions using the DE algorithm to optimize a single pair of globally shared fusion weights. Compared with the individual backbone models, their ensemble achieved improved classification performance across disease categories. However, the proposed framework primarily applies DE as a black-box optimizer for model-level fusion and relies on globally shared fusion coefficients. Unlike Li et al., our study presents a comprehensive methodological framework that includes a complete mathematical formulation of the

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DE optimization framework, including the optimization objective, mutation, crossover, selection operators, optimization equations, algorithmic pseudocode, and computational complexity analysis. Furthermore, our work extends the conventional DE-based global fusion framework to a disease-adaptive class-specific fusion strategy. Rather than only optimizing ensemble weights, we investigate a class-specific fusion strategy that learns independent fusion coefficients for each disease category. We systematically evaluate multiple CNN–Transformer backbone combinations, compare the proposed DE-based fusion frameworks with several representative decision-level fusion strategies, and support the experimental findings through bootstrap confidence intervals and statistical significance analysis.

Recent studies have further explored structured feature integration and class-aware representation learning. XProtoNet [19] introduces an occurrence-based weighting mechanism to improve interpretability in CXR diagnosis. By incorporating dynamic patch selection, the model adapts to variations in lesion size and spatial distribution. Similarly, GLoRI integrates Transformer representations with patch-level CNN features to improve disease classification [20]. The model employs class-aware representation integration to enhance feature learning across large-scale CXR datasets. Although these approaches advance feature-level integration and class-aware representation learning, integration remains primarily at the feature level, without explicitly optimizing class-specific fusion at the decision stage.

III. VISUAL CHARACTERISTICS OF THORACIC DISEASES

Prior studies on the NIH CXR dataset [9] [12] [21] [22] [23] have shown that different disease categories exhibit distinct spatial characteristics, with certain pathologies presenting localized findings and others demonstrating more diffuse or globally distributed patterns. For instance, cardiomegaly is defined based on the cardiothoracic ratio, which reflects the size of the heart relative to the thoracic cage; therefore, evaluation is performed at the organ level [24]. Pneumonia can appear on a CXR either as a localized area of increased opacity in one region or as a more diffuse pattern involving larger regions of the lung [25]. Pneumothorax is most often identified on a CXR by a thin line along the lung margin and the loss of normal lung markings beyond that line; in some cases, especially in a supine patient, air may be seen more prominently at the lung base or adjacent to the heart. Therefore, the assessment is primarily based on local findings; however, the overall context, such as the patient's position and underlying lung disease, must always be taken into account [26]. A pulmonary nodule appears as a small, well-defined opacity within the lung tissue. Its evaluation mainly relies on local imaging characteristics such as size, location, border appearance, and density. When follow-up imaging is available, temporal changes such as stability or growth also provide important diagnostic information. Lesions larger than 3 cm in diameter are classified as pulmonary masses [27]. Similar to nodules, pulmonary masses are assessed primarily based on their shape, margin, and internal structure properties. Therefore, both nodules and masses are considered predominantly associated with local visual manifestations. Atelectasis is typically seen on a CXR as a region of increased opacity, often located near the lung periphery. The borders are usually ill-defined and may appear contiguous with thickened pleura. Mild volume loss may also be present, and in some cases, fine linear streaks converging toward the affected area can be observed. Therefore, evaluation is primarily based on these localized imaging findings [28]. Infiltration refers to diffuse opacities distributed across the lung fields, often reflecting interstitial or inflammatory processes. Since evaluation depends on the extent and spatial spread of these opacities, infiltration tends to exhibit more globally distributed radiographic patterns. Pleural effusion is typically characterized by fluid accumulation in the pleural space, presenting as diffuse opacity and blunting of the costophrenic angles. Its assessment relies on the distribution and volume of fluid across the thoracic cavity [29].

IV. METHOD

A. DATASET

We trained and evaluated the proposed framework on the NIH CXR dataset, which consists of 112,120 frontal-view CXR collected from 30,805 patients [9]. The dataset provides image-level disease annotations extracted from associated radiology reports using natural language processing techniques. In the study, we focus on eight thoracic disease categories: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, and Pneumothorax (Table 1). These diseases exhibit diverse spatial characteristics, making the NIH CXR dataset suitable for evaluating class-specific decision-level fusion methods across multiple thoracic disease categories.

TABLE 1. Class distribution and localization subset statistics of the NIH CXR dataset

Class	Positive Samples	Negative Samples	Positive Ratio(%)	BBox Samples
Nodule	5193	32827	13.66	68
Atelectasis	9187	28833	24.16	153
Mass	4414	33606	11.61	59
Pneumothorax	4049	33971	10.65	73
Pneumonia	927	37093	2.44	78
Effusion	10306	27714	27.11	116
Infiltration	16421	21599	43.19	94
Cardiomegaly	2314	35706	6.09	122

The dataset includes a subset with bounding box annotations that indicate the location of lesions. Class-wise statistics of this subset are given in the last column of Table 4. These annotations were used to analyze how diseases are distributed spatially and to compute the Local–Global Index (Section V-A).

B. BASE MODELS

1) CNN-based Model

Convolutional Neural Networks (CNNs) are widely used deep learning architectures for image analysis that learn hierarchical feature representations through stacked convolutional lay-

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ers [30]. By applying convolutional filters over small spatial regions of an image, CNNs effectively capture fine-grained visual patterns such as edges, textures, and small localized structures. This design makes CNNs particularly effective for detecting subtle radiographic patterns in CXRs, including small nodules, focal opacities, and texture-level abnormalities. In this study, we use DenseNet [31] and MobileNet [32] as the base CNN architectures. DenseNet is selected because its dense connectivity pattern allows features to be shared across layers and helps the network learn informative representations from CXR images. MobileNet is incorporated as a structurally different lightweight CNN architecture. Its depthwise separable convolutional design reduces the number of parameters and computational cost while preserving competitive representation capability.

2) Transformer-based Model

ViT models images as sequences of patches and learns relationships between distant regions through self-attention mechanisms [33], [34]. By allowing each patch to attend to all others, ViT effectively learns global contextual information and spatial relationships across the entire image. This capability makes transformers suitable for medical imaging tasks in which disease patterns are distributed over large anatomical regions rather than confined to localized structures. Despite these advantages, standard ViT architectures have limited ability to capture fine-grained local details and lack an inherent hierarchical feature representation [35]. As a result, their effectiveness may be reduced for detecting subtle radiographic abnormalities that rely on detailed textures, edge-level cues, or small localized patterns.

The Swin Transformer [7] partitions the input image into non-overlapping local windows and computes self-attention only within each window, which reduces computational cost while capturing local texture and edge information. To allow information exchange across windows, the window partitioning scheme is shifted between consecutive layers. As a result, image patches interact with different neighboring regions across layers, gradually capturing global contextual relationships.

C. PREPROCESSING STAGE

1) Preprocessing of CXR

Before feeding the CXR images into the base models, preprocessing steps were applied to improve the prediction performance of the models. Due to the imbalanced distribution of disease categories, data augmentation techniques were used to increase the representation of underrepresented classes. Random rotations, translations, and zoom operations were applied to the training CXR images. To improve the discrimination performance of visual clues, Contrast Limited Adaptive Histogram Equalization (CLAHE) [36] was applied to each CXR. CLAHE enhances local contrast and highlights subtle intensity variations, thereby improving the visibility of fine structural details. Subsequently, lung regions were segmented using a U-Net–based architecture [37]. U-Net was selected because of its strong performance in biomedical image segmentation tasks with limited annotated data. The network have an encoder–decoder architecture with skip connections that preserve spatial information and improve segmentation accuracy. The segmentation model was trained on the Japanese Society of Radiological Technology dataset [42], which contains expert-annotated lung masks for CXR [43]. The resulting lung masks were used to remove non-lung regions, thereby focusing the analysis on clinically relevant pulmonary anatomy. Finally, all segmented images were resized to a fixed resolution and normalized to ensure consistent input distributions for model training.

2) Probability Calibration Using Isotonic Regression

Before applying the late fusion methods, the prediction probabilities produced by the CNN- and Transformer-based models were calibrated using Isotonic Regression [44], [45]. Deep learning models may produce probability estimates that are not well calibrated, even when their classification accuracy is high. Since the late fusion methods directly combine prediction probabilities, the quality of these probabilities directly affects the fusion process. If the predicted probabilities are overconfident or underconfident, the fusion method may assign suboptimal fusion weights to the individual models. Probability calibration provides more reliable confidence estimates, allowing the fusion methods to learn more effective fusion weights and improving ensemble performance. Because the CNN and Transformer models exhibited different probability distributions, a separate calibrator was trained for each model. Furthermore, since the problem has a multi-label classification structure, each disease category was treated as an independent binary classification task, and a separate Isotonic Regression model was trained for each model–class pair. The calibrated probabilities were then used as inputs to all late fusion methods.

D. DIFFERENTIAL EVOLUTION-BASED ENSEMBLE LEARNING

To effectively integrate the complementary strengths of CNN- and Transformer-based architectures, we use the Differential Evolution (DE) algorithm [38] as an optimization framework for ensemble fusion. The coefficients are optimized under two different ensemble formulations: (i) DE-based global fusion, in which a single pair of globally shared fusion weights is learned to combine CNN and Transformer predictions across all disease categories, and (ii) DE-based class-specific fusion, a disease-aware ensemble strategy in which independent fusion weights are assigned to each disease class. DE is a population-based derivative-free optimization algorithm that efficiently searches continuous parameter spaces without requiring gradient information, making it well suited for optimizing nonlinear objective functions through global exploration of the search space.

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1) Differential Evolution-Based Global Fusion

Let p^{CNN} and p^{ViT} denote the predicted probabilities of the CNN-based and Transformer-based models. The ensemble prediction is computed as:

$$p^{Ensemble} = w^{CNN}\, p^{CNN} + w^{ViT}\, p^{ViT} \tag{1}$$

where $w^{CNN}, w^{ViT} \in [0,1]$ are globally shared fusion coefficients satisfying $w^{CNN} + w^{ViT} = 1$. The fusion weights are optimized using DE, an evolutionary optimization algorithm that iteratively searches for optimal solutions through mutation, crossover, and selection operations. Each individual $\mathbf{x} = [w^{CNN}, w^{ViT}] \in \mathbb{R}^2$ in the DE population represents a candidate solution corresponding to the fusion weights. At each generation G, the algorithm generates new candidates by modifying existing weight vectors using the differences between randomly selected individuals in the population.

$$v_{i,G+1} = x_{r_1,G} + F\left(x_{r_2,G} - x_{r_3,G}\right), \tag{2}$$

where $x_{r_1,G}$, $x_{r_2,G}$ and $x_{r_3,G}$ are randomly selected weight vectors from the population, and $F \in (0,2)$ is the scaling factor. In the mutation step, a new candidate solution (mutant vector) is generated by combining information from the weight vectors in the current population. Specifically, the difference between two randomly selected individuals $(x_{r_2,G} - x_{r_3,G})$ represents a directional variation in the search space, which is then scaled by the factor F to control the magnitude of the update. This difference is added to a third individual $x_{r_1,G}$, resulting in a new candidate vector $v_{i,G+1}$.

In the crossover stage, a trial vector is generated by combining the mutant vector $v_{i,G+1}$ and the current solution $x_{i,G}$. For each component j, the value is selected either from the mutant vector or from the current solution based on a probabilistic rule controlled by the crossover rate CR. This mechanism enables the algorithm to combine newly generated candidate vectors with the current solution, allowing it to generate new solutions while maintaining and improving previously found ones.

$$u_{j,i,G+1} = \begin{cases} v_{j,i,G+1}, & \text{if } \mathrm{randb}(j) \le CR \text{ or } j = \mathrm{rnbr}(i), \\ x_{j,i,G}, & \text{otherwise,} \end{cases} \tag{3}$$

where $CR \in [0,1]$ denotes the crossover rate and $\mathrm{randb}(j) \in [0,1]$ is a random number generated independently for each component j. If $\mathrm{randb}(j) \le CR$, the j-th component is taken from the mutant vector; otherwise, it is taken from the current solution. The function $\mathrm{rnbr}(i)$ ensures that at least one component is inherited from the mutant vector, preventing the trial vector from being identical to the current solution $x_{i,G}$. Here, $u_{i,G+1} \in \mathbb{R}^2$ denotes the trial vector that will be evaluated in the selection stage.

In the selection stage, the trial vector is compared with the current solution, and the solution with better performance is selected for the next generation.

$$x_{i,G+1} = \begin{cases} u_{i,G+1}, & \text{if } f(u_{i,G+1}) \ge f(x_{i,G}), \\ x_{i,G}, & \text{otherwise.} \end{cases} \tag{4}$$

Algorithm 1 Pseudocode of Differential Evolution for Optimizing Fusion Weights

```
Initialize population x_i, i = 1, ..., N
for each generation g = 1 to G do
    for each individual x_i do
        Randomly select x_r1, x_r2, x_r3
        Mutation: v_i = x_r1 + F(x_r2 − x_r3)
        Crossover:
        Generate random index rnbr(i) ∈ {1, ..., d}
        for j = 1 to d do
            if randb ≤ CR or j = rnbr(i) then
                u_i[j] = v_i[j]
            else
                u_i[j] = x_i[j]
            end if
        end for
        Evaluate fitness f(u_i)
        Selection:
        if f(u_i) > f(x_i) then
            x_i = u_i
        end if
    end for
end for
return best solution
```

where $f(\cdot)$ denotes the fitness (performance evaluation) function. The trial vector $u_{i,G+1}$, generated through mutation and crossover, is evaluated using the fitness function. If it produces better performance than the current vector $x_{i,G}$, it proceeds to the next generation; otherwise, the current solution is retained. The fitness function is defined as the macro-average AUC corresponding to a given set of globally shared fusion weights. Algorithm 1 summarizes the optimization process for learning the fusion weights.

2) Differential Evolution-Based Class-Specific Fusion

In the second fusion strategy, we adopt a class-specific ensemble learning framework in which independent fusion weights are assigned to each disease category. Let p_c^{CNN} and p_c^{ViT} denote the predicted probabilities produced by the CNN-based and Transformer-based models for disease class $c \in \{1,\ldots,C\}$. The ensemble prediction for class c is computed as:

$$p_c^{\mathrm{Ensemble}} = w_c^{\mathrm{CNN}}\, p_c^{\mathrm{CNN}} + w_c^{\mathrm{ViT}}\, p_c^{\mathrm{ViT}} \tag{5}$$

where w_c^{CNN} and $w_c^{\mathrm{ViT}} \in [0,1]$ are class-specific fusion coefficients, with $w_c^{\mathrm{CNN}} + w_c^{\mathrm{ViT}} = 1$ as a constraint. The class-specific fusion weights are optimized using the differential evolution algorithm. The optimization process searches for the optimal weight matrix:

$$W = \begin{bmatrix} w_1^{\mathrm{CNN}} & w_1^{\mathrm{ViT}} \\ w_2^{\mathrm{CNN}} & w_2^{\mathrm{ViT}} \\ \vdots & \vdots \\ w_C^{\mathrm{CNN}} & w_C^{\mathrm{ViT}} \end{bmatrix} \in \mathbb{R}^{C\times 2} \tag{6}$$

which holds the relative contributions of each architecture for every disease class. Each individual in the DE population represents a candidate solution corresponding to the vectorized form of the weight matrix W. The matrix is flattened

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into a vector $\mathbf{x} \in \mathbb{R}^{2C}$ prior to optimization. The same DE optimization procedure described in Section IV-D1 is applied to optimize the class-specific weight matrix W. Although each disease class has its own fusion weights, the DE-CSF framework optimizes all class-specific weights simultaneously using the macro-average AUC as the objective function. This unified optimization framework avoids the need for multiple independent optimization runs, thereby simplifying the overall optimization process.

DE-CSF introduces a higher computational cost than DE-GF due to the increased number of independent optimization variables. Because the fusion weights have the constraints $w_c^{CNN}, w_c^{ViT} \in [0,1]$ and $w_c^{CNN} + w_c^{ViT} = 1$, only one independent variable is optimized for each disease class. Accordingly, DE-GF optimizes one globally shared fusion weight, whereas DE-CSF optimizes C class-specific fusion weights (8 independent parameters for $C = 8$ classes in our study). In our implementation, however, two raw fusion weights were initially defined for each disease class, resulting in a total of 16 raw weights. The Differential Evolution algorithm optimized these raw weights under the bound constraint $[0,1]$ specified through the *bounds* parameter. After each optimization step, the two raw weights corresponding to each disease class were normalized by their sum, ensuring that the final fusion weights remained within $[0,1]$ and satisfied $w_c^{CNN} + w_c^{ViT} = 1$. Therefore, although 16 raw weights were used internally during the optimization process, they correspond to eight independent optimization variables. The complexity of the DE optimization is $\mathcal{O}(P \cdot G \cdot C)$, where P is the population size factor, G is the number of generations, and C is the number of disease classes (Table 2). DE-CSF scales linearly with the number of classes. However, the DE optimization operates only on the predicted probabilities of pre-trained models and does not involve backpropagation or retraining of the backbone networks. Therefore, the additional cost introduced by DE-CSF remains limited to a lightweight post-training optimization step.

TABLE 2. Computational cost analysis of DE-GF and DE-CSF fusion strategies

Metric	DE-GF	DE-CSF
Independent Optimized param. (C)	1	8
Population size factor (P)	50	50
Maximum generations (G)	100	100

V. EXPERIMENTS

A. EMPIRICAL ANALYSIS OF LOCAL AND GLOBAL DISEASE CHARACTERISTICS

To provide empirical insight into the spatial characteristics of thoracic diseases, we quantified whether each pathology exhibits local abnormalities or more globally distributed patterns using a Local–Global Index. The index was computed based on bounding box annotations provided in the NIH CXR dataset. Each bounding box localizes the diseased region within the CXR. Using this region-level information, we measured the geometric coverage of disease manifestations. For each bounding box, we calculated the relative area, the horizontal coverage ratio and vertical coverage ratio as follows:

$$\text{Area ratio:} \quad r = \frac{\sum_k A_k}{A_I} \tag{7}$$

$$\text{Width coverage:} \quad r_w = \frac{\sum_k w_k}{W_I} \tag{8}$$

$$\text{Height coverage:} \quad r_h = \frac{\sum_k h_k}{H_I} \tag{9}$$

where k indexes the bounding-box region within an image I; A_k denotes the area of the k-th region; A_I denotes the total image area; and W_I and H_I denote the image width and height, respectively. These geometric measures were then combined into a score (Eq. 10), which reflects how broadly a disease region spans the image.

$$\text{Global score:} \quad s = r + r_w + r_h \tag{10}$$

For each annotated region, the score s was computed to determine whether the disease manifestation occupies a localized region or extends broadly across the thoracic cavity. The scores were thresholded using a threshold value τ to assign each region a local or global label ℓ_i according to Eq. 11.

$$\ell_i = \begin{cases} \text{global}, & s_i \geq \tau, \\ \text{local}, & s_i < \tau, \end{cases} \qquad \tau = T\left(\{s_i\}_{i=1}^{N}\right) \tag{11}$$

where $T(\cdot)$ denotes a thresholding function applied to the distribution of Local-Global Index scores. These region-level labels were subsequently aggregated at the class level to estimate the proportion of local and global manifestations for each pathology.

To assess the robustness of the Local-Global Index, multiple thresholding strategies were evaluated, including the mean, median, K-means clustering, Huber M-estimation, and Gaussian Mixture Model (GMM)-based approaches. K-means determines the threshold by partitioning the score distribution into two clusters, Huber provides a robust threshold that is less sensitive to outliers, and GMM estimates the threshold by modeling the score distribution as a mixture of two Gaussian components.

TABLE 3. Percentage of samples classified as global-dominant for each disease category under different Local–Global Index threshold definitions.

Class	n	Mean	Median	K-means	Huber	GMM
Nodule	68	0.00	0.00	0.00	0.00	2.63
Atelectasis	153	11.76	21.57	9.80	11.76	16.14
Mass	59	18.64	23.73	15.25	22.03	22.17
Pneumothorax	73	36.99	49.32	26.03	39.73	37.83
Pneumonia	78	38.46	65.38	21.79	41.03	39.52
Effusion	116	40.52	56.03	31.90	43.10	43.35
Infiltration	94	43.62	64.89	31.91	45.74	46.10
Cardiomegaly	122	100.00	100.00	100.00	100.00	99.89

Table 3 summarizes the percentage of samples classified as global-dominant under each threshold definition. Although the absolute percentages vary across thresholding strategies,

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the overall ordering of disease categories and their broad spatial characterization remain largely consistent. The analyzed samples can be broadly grouped into four spatial tendency categories: (i) strongly local (Nodule), (ii) predominantly local (Atelectasis and Mass), (iii) diseases with intermediate local–global distributions (Pneumothorax, Pneumonia, Effusion, and Infiltration), and (iv) strongly global (Cardiomegaly). The observed local–global distribution patterns are consistent with the radiological characteristics of thoracic diseases discussed in Section III (Visual Characteristics of Thoracic Diseases). Pathologies clinically defined by localized and spatially confined abnormalities, such as nodules, masses and atelectatic regions, exhibit predominantly localized spatial patterns in the empirical analysis. Conversely, diseases associated with organ-level enlargement or diffuse pulmonary involvement such as cardiomegaly, infiltration, and pleural effusion, demonstrate a predominantly global distribution. Findings such as pneumonia and pneumothorax display more balanced spatial characteristics, reflecting their potential to manifest both as localized abnormalities and as broader thoracic changes depending on disease severity. However, the number of available bounding-box annotations is relatively limited for certain disease categories, particularly Nodule, Mass, and Pneumothorax. Therefore, the reported local–global tendencies for these classes should be interpreted as indicative observations rather than definitive disease-specific conclusions.

B. EXPERIMENTAL SETUP AND BENCHMARK MODELS

The experiments were conducted on the NIH CXR dataset. The dataset was randomly divided at the patient level into training, validation, and test sets with proportions of 70%, 15%, and 15%, respectively, ensuring that data from the same patient appeared in only one subset. The class-wise sample distributions across the training, validation, and test sets are presented in Table 4. To avoid data leakage, the training, hyperparameter tuning, probability calibration, fusion optimization, and final testing stages were performed using separated data subsets. The base models were trained using only the training set. Hyperparameter tuning, checkpoint selection, isotonic probability calibration, and DE-based fusion weight optimization were conducted on the validation set. For the classifier-based fusion methods, the validation set was further split into 80% training and 20% validation subsets for fusion-model training. Subsequently, all resulting models were evaluated on the independent test dataset. Due to the imbalanced distribution of disease categories in the dataset, data augmentation techniques were used to increase the representation of underrepresented classes. The augmentation operations and their corresponding parameter settings are listed in Table 5. DenseNet121 and MobileNetV3 were initialized using the pretrained models available in the TensorFlow/Keras Applications library, while ViT and Swin-T were initialized from publicly available ImageNet pretrained checkpoints. As Transformer backbones, we use the ViT-Base/16 and Swin Transformer V2 Tiny 256 models. All models were subsequently fine-tuned on the NIH CXR training set. The models were implemented in Python using TensorFlow–Keras. Training was performed on a workstation equipped with an NVIDIA RTX 4090 GPU (24 GB VRAM) with CUDA v11 support.

TABLE 4. Class distribution across the training (70%), validation (15%), and test (15%) datasets

Class	Train	Val	Test	Total
Nodule	3640	709	844	5193
Atelectasis	6473	1398	1316	9187
Mass	2970	735	709	4414
Pneumothorax	2844	549	656	4049
Pneumonia	680	135	112	927
Effusion	7071	1680	1555	10306
Infiltration	11590	2422	2409	16421
Cardiomegaly	1597	327	390	2314

TABLE 5. Data augmentation parameters applied to the CXR images.

Augmentation	Parameter
Rotation	$\pm 5^{\circ}$
Width translation	$\pm 5\%$
Width zoom	0.9–1.1×
Height zoom	0.9–1.1×
Fill mode	Nearest

To evaluate the effectiveness of the DE-based fusion strategies, we compared them against several baseline and ensemble models. As baseline models, we used CNN-based architectures (DenseNet and MobileNet) and Transformer-based architectures (standard ViT and Swin Transformer). For ensemble methods, we implemented Ensemble-Max, Ensemble-MLP, Ensemble-Soft Gating and Grid-CSF fusion strategies. Ensemble-Max is a heuristic approach that combines CNN and Transformer predictions using a maximum operation without learnable fusion weights. Ensemble-MLP performs late fusion by feeding the predicted probabilities of the base models into a multilayer perceptron (MLP) to learn nonlinear combinations. For the Ensemble-MLP model, two architectural variants with one and two hidden layers were evaluated to assess the effect of model capacity on fusion performance. Ensemble-Soft Gating is a fusion strategy in which model outputs are combined through a gating network. The output vectors produced by the base models are fed into a small gating network as input. This network generates soft weights representing the contribution each model for each sample [39]. To provide an alternative class-specific optimization approach, the Grid-CSF method was implemented. Grid-CSF adopts a class-specific weighting strategy by determining separate fusion weights for each disease class. The fusion weights are optimized using a grid-search procedure, which systematically evaluates candidate weight combinations within a predefined discrete search space for each disease class. Two DE-based fusion variants were evaluated: (i) DE-GF (Global Fusion), which learns a single pair of globally shared fusion weights for CNN and Transformer predictions; and (ii) DE-CSF (Class-Specific Fusion), which learns independent fusion weights for each disease class.

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The hyperparameters of the base models were selected using a random search strategy (Table 6). The search was conducted over predefined candidate values, and each hyperparameter combination was evaluated based on the validation AUC. The combination achieving the highest validation AUC was selected as the final model configuration. The selected hyperparameters for the base models and fusion methods are summarized in Tables 7 and 8, respectively.

TABLE 6. Hyperparameter search spaces of the base models

Model	Hyperparameter	Search Space
CNN-based	Dense 1	32, 64, 128, 256, 512
	Dense 2	16, 32, 64, 128, 256
	Dropout rate	0.3, 0.4, 0.5
	L2 rate	0.0001, 0.0003, 0.0005
	Focal loss α	0.25, 0.50, 0.75
	Focal loss γ	1.0, 2.5, 4.5
	Learning rate	1×10^{-5}, 1×10^{-4}, 2×10^{-4}, 3×10^{-4}, 5×10^{-4}, 1×10^{-3}
Transformer-based	Dense 1	64, 128, 256, 512
	Dense 2	32, 64, 128, 256, 512
	Dropout rate	0.3, 0.4, 0.5
	L2 rate	0.0001, 0.0003, 0.0005
	Focal loss α	0.25, 0.5, 0.75, 1.0
	Focal loss γ	0.5, 1.0, 2.0, 2.5, 3.5, 4.5
	Learning rate	1×10^{-5}, 1×10^{-4}, 2×10^{-4}, 3×10^{-4}, 5×10^{-4}, 1×10^{-3}
	Weight decay	0.0005 0.0001, 0.001, 0.01

The models were trained using a hybrid loss function that combines Focal Loss and Asymmetric Loss (ASL) [17]. The two loss components were integrated through a Pareto-inspired dynamic weighting strategy based on their relative loss magnitudes. The random seeds were fixed at 42. The same seed value was used for both dataset splitting and model training. Model checkpoint selection was based on the AUC achieved on the validation set. The validation AUC was monitored at the end of each epoch, and the weights of the model achieving the highest validation AUC were saved as the final checkpoint.

C. PERFORMANCE OF DE-BASED FUSION STRATEGIES

1) Overall Performance Comparison

The performance of the DE-based fusion strategies is evaluated using the Area Under the ROC Curve (AUC) and the Area Under the Precision–Recall Curve (AUPRC), computed separately for each disease category. Table 9 presents a comparison of baseline models, ensemble methods, and the DE-based fusion strategies. The table reports class-wise AUC results for different backbone configurations, with results presented as mean values with 95% confidence intervals. In addition to reporting performance metrics, we assess the statistical significance of the observed differences. Statistical comparisons between the proposed DE-CSF method and each competing approach were performed using a paired permutation test, which preserves the patient-level pairing between model predictions while requiring no distributional assumptions. During each permutation, the paired model predictions for each patient were randomly exchanged between the two competing models, and the evaluation metric was recomputed. The Holm correction was applied to the resulting p-values to account for multiple comparisons, and both the raw and Holm-adjusted p-values are reported in Table 9. We also quantify performance improvements using the mean bootstrap AUC difference (ΔAUC), which represents the average paired difference between the AUC values of the reference

TABLE 7. Training hyperparameters of base models

Model	Hyperparameter	Value
DenseNet	Input/output layers	$224 \times 224 \times 3$ / 8
	Dense 1	128
	Dense 2	64
	Dropout rate	0.3
	L2 rate	0.0005
	Learning rate	0.001
	Batch size	128
	Epochs	50
	Activation f. (hidden)	ReLU
	Activation f. (output)	Sigmoid
	Optimizer	Adam
	Loss function (hybrid)	Focal loss + ASL
	Focal loss α	0.25
	Focal loss γ	4.5
	ASL γ_+	1.0
	ASL γ_-	4.0
MobileNet	Input/output layers	$224 \times 224 \times 3$ / 8
	Dense 1	128
	Dense 2	32
	Dropout rate	0.5
	Weight decay	0.0001
	Learning rate	0.0001
	Batch size	128
	Epochs	50
	Activation f. (hidden)	GeLU
	Activation f. (output)	Sigmoid
	Optimizer	Adam
	Loss function (hybrid)	Focal loss + ASL
	Focal loss α	0.5
	Focal loss γ	1.0
	ASL γ_+	1.5
	ASL γ_-	4.0
ViT	Input/output layers	$224 \times 224 \times 3$ / 8
	Dense 1	512
	Dense 2	512
	Dropout rate	0.3
	L2 rate	0.0001
	Weight decay	0.0005
	Learning rate	0.0003
	Batch size	64
	Epochs	50
	Activation f. (hidden)	GeLU
	Activation f. (output)	Sigmoid
	Optimizer	AdamW
	Loss function (hybrid)	Focal loss + ASL
	Focal loss α	1.0
	Focal loss γ	3.5
	ASL γ_+	1.0
	ASL γ_-	4.0
Swin Transformer	Input/output layers	$256 \times 256 \times 3$ / 8
	Dense	512
	Dropout rate	0.3
	L2 rate	0.0001
	Weight decay	0.0001
	Learning rate	0.0001
	Batch size	64
	Epochs	50
	Activation f. (hidden)	GeLU
	Activation f. (output)	Sigmoid
	Optimizer	AdamW
	Loss function (hybrid)	Focal loss + ASL
	Focal loss α	0.75
	Focal loss γ	2.0
	ASL γ_+	1.0
	ASL γ_-	4.0

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TABLE 8. Training hyperparameters for fusion methods

Method	Hyperparameter	Value
Ensemble-MLP	Input/output layers	16 / 8
	Dense 1	256
	Dropout rate	0.3
	L2 rate	0.0003
	Learning rate	0.0005
	Batch size	128
	Epochs	100
	Activation f. (hidden)	ReLU
	Activation f. (output)	Sigmoid
	Optimizer	AdamW
	Loss function (hybrid)	Focal loss + ASL
	Focal loss α	0.75
	Focal loss γ	0.5
	ASL γ_+	1.0
	ASL γ_-	4.0
Ensemble-Soft Gating	Input/output layers	8 + 8 / 8
	Dense 1	32
	L2 rate	0.001
	Learning rate	0.001
	Batch size	128
	Epochs	50
	Activation f. (hidden)	ReLU
	Activation f. (gate)	Softmax
	Optimizer	AdamW
	Loss function (hybrid)	Focal loss + ASL
	Focal loss α	0.75
	Focal loss γ	0.5
	ASL γ_+	1.0
	ASL γ_-	4.0
Grid-CSF	Search method	Grid search
	Weight bounds	$[0, 1]$
	Step size	0.01
DE-GF	Generations	100
	Tolerance	1×10^{-4}
	Population size	50
	Mutation factor (F)	(0.5, 1)
	Crossover rate (CR)	0.7
	Polish	True
	Weight bounds	$[0, 1]$
DE-CSF	Generations	100
	Tolerance	1×10^{-4}
	Population size	50
	Mutation factor (F)	$(0.5, 1)$
	Crossover rate (CR)	0.7
	Polish	True
	Weight bounds	$[0, 1]$

and compared models across all bootstrap samples. The mean paired bootstrap difference was computed as follows:

$$\Delta\mathrm{AUC} = \frac{1}{B}\sum_{b=1}^{B}\left(\mathrm{AUC}_{\mathrm{ref}}^{(b)} - \mathrm{AUC}_{\mathrm{cmp}}^{(b)}\right),$$

where B denotes the total number of bootstrap resamples, and $\mathrm{AUC}_{\mathrm{ref}}^{(b)}$ and $\mathrm{AUC}_{\mathrm{cmp}}^{(b)}$ represent the AUC values of the reference and compared models obtained from the b-th bootstrap resample, respectively.

Table 9 is organized into distinct comparison blocks. The first four blocks present results obtained under our experimental protocol using different CNN–Transformer backbone combinations and fusion strategies. The first block presents results based on the DenseNet and ViT backbones. In this configuration, each backbone is first evaluated individually to establish baseline performance. Their prediction outputs are then combined using conventional decision-level fusion strategies (Ensemble-Max, Ensemble-MLP with one- and two-layer architectures, and Ensemble-Soft Gating), the Grid-CSF, and the DE-based fusion approaches (DE-GF and DE-CSF). This setup provides a comparison between heuristic, learnable, and optimization-based fusion strategies. The second block follows the same evaluation protocol as the first block, but replaces the DenseNet backbone with MobileNet. This allows us to evaluate the DE-based fusion strategies using an architecture with different computational characteristics. This demonstrates the robustness of the proposed method across different CNN architectures. The third and fourth blocks follow the same evaluation protocol but replace the ViT backbone with the Swin Transformer. In these settings, CNN backbones are combined with a hybrid Transformer architecture that inherently captures both local and global dependencies. The resulting fusion experiments demonstrate how the DE-based strategies perform when applied to stronger backbone combinations. The fifth block reports previously published results on the NIH CXR dataset. These methods were not reproduced within our experimental framework and are included only as non-equivalent reference results. Accordingly, this section of the table is not intended for direct quantitative comparison with the proposed method or the benchmark models evaluated under our experimental protocol. Specifically, the findings for ResNet50, SEResNet50, Conformer, ConvNeXt, RepVGG, DeiT, and CTransCNN are taken from [17], while the results for DualCheXN [40], ImageGCN [41], and CheX-DS [13] are obtained from their respective original studies. Due to differences in dataset composition, preprocessing pipelines, disease categories included in evaluation, and evaluation protocols, these literature-based results should not be interpreted as direct comparisons with the proposed approach. Instead, they are provided solely to contextualize the range of performances previously reported on the NIH CXR dataset.

The results in Table 9 indicate that CNN- and Transformer-based architectures exhibit complementary representational strengths. Across all backbone combinations, the Transformer-based models consistently achieve higher mean AUC values than their CNN counterparts. Combining the predictions of the two architectures further improves performance with all ensemble strategies, outperforming the corresponding individual backbones. Among the evaluated fusion methods, Ensemble-Soft Gating, Grid-CSF, DE-GF, and DE-CSF consistently achieve the highest performance across both individual disease categories and overall. Among these methods, DE-CSF yields the highest mean AUC for all four CNN–Transformer backbone combinations, while Grid-CSF consistently performs slightly better than DE-GF and remains highly competitive with DE-CSF. These results indicate that class-specific fusion provides a systematic advantage over globally shared fusion, whereas the proposed DE-based optimization further improves performance through optimization of the fusion weights. The resulting confidence intervals, p-values, and bootstrap difference analysis indicate that the improvements achieved by the DE-based fusion strategies are consistently observed across bootstrap samples. Although the absolute performance differences between DE-CSF and DE-GF are relatively small in terms of mean AUC, these

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TABLE 9. Comparison of class-wise and mean AUC (%) performance of baseline models, conventional ensemble methods, and DE-based fusion strategies on the NIH CXR dataset.

Obtained Under Our Experimental Protocol												
Model	**Nodule**	**Atelectasis**	**Mass**	**Pneumothorax**	**Effusion**	**Infiltration**	**Pneumonia**	**Cardiomegaly**	**Mean**	**p-value / Holm**	**ΔAUC**
Base: DenseNet	72.75 [70.59, 74.81]	75.18 [73.42, 76.97]	76.99 [74.43, 79.52]	81.19 [78.82, 83.55]	81.48 [79.93, 82.95]	68.79 [67.03, 70.43]	60.54 [55.99, 65.01]	89.80 [87.49, 91.92]	75.84 [74.87, 76.74]	1E−04 / 8E−04	3.31 [2.74, 3.94]
Base: ViT	73.44 [71.22, 75.68]	76.92 [75.23, 78.57]	77.59 [74.96, 79.92]	83.19 [80.55, 85.38]	82.57 [80.98, 84.05]	68.94 [67.30, 70.58]	60.76 [55.21, 65.82]	90.68 [88.67, 92.38]	76.76 [75.74, 77.81]	1E−04 / 8E−04	2.40 [1.86, 2.93]
Ensemble-MLP	74.32 [72.01, 76.45]	77.40 [75.71, 79.03]	78.44 [76.01, 80.76]	83.76 [81.35, 86.15]	84.08 [82.58, 85.52]	**71.01 [69.37, 72.60]**	58.63 [53.12, 64.80]	90.54 [88.01, 92.66]	77.27 [76.18, 78.36]	8E−03 / 28E−03	1.86 [0.99, 2.67]
Ensemble-MLP (Two Layer)	75.37 [73.15, 77.45]	77.58 [76.01, 79.20]	78.78 [76.32, 81.19]	83.68 [81.23, 85.99]	84.40 [82.87, 85.81]	70.78 [69.17, 72.38]	58.80 [53.31, 64.31]	91.10 [88.88, 93.04]	77.56 [76.54, 78.56]	2E−02 / 28E−03	1.57 [0.82, 2.34]
Ensemble-Max	75.30 [73.12, 77.27]	77.21 [75.58, 78.85]	79.44 [77.08, 81.74]	83.75 [81.43, 85.83]	83.66 [82.19, 85.11]	70.22 [68.55, 71.77]	62.84 [57.64, 67.79]	90.94 [88.84, 92.76]	77.92 [76.90, 78.92]	1E−04 / 8E−04	1.22 [0.96, 1.48]
Ensemble-Soft Gating	76.01 [73.87, 77.96]	77.66 [76.00, 79.30]	79.96 [77.60, 82.31]	84.46 [82.18, 86.58]	83.92 [82.47, 85.30]	70.61 [68.96, 72.15]	64.24 [58.92, 69.26]	91.57 [89.50, 93.32]	78.55 [77.58, 79.50]	8E−04 / 4E−03	0.59 [0.36, 0.81]
DE-GF	76.12 [73.99, 78.09]	77.88 [76.22, 79.52]	80.06 [77.69, 82.37]	84.64 [82.32, 86.75]	84.14 [82.65, 85.53]	70.71 [69.08, 72.27]	64.74 [59.50, 69.85]	91.63 [89.60, 93.38]	78.74 [77.73, 79.71]	7E−03 / 28E−03	0.41 [0.26, 0.56]
Grid-GSF	76.10 [73.96, 78.09]	77.93 [76.28, 79.53]	80.06 [77.63, 82.36]	84.73 [82.40, 86.82]	84.14 [82.65, 85.53]	70.72 [69.08, 72.26]	64.77 [59.45, 66.71]	91.64 [89.61, 93.35]	78.76 [77.77, 79.76]	7E−03 / 28E−03	0.39 [0.25, 0.53]
DE-CSF	**76.48 [74.41, 78.42]**	**78.25 [76.64, 79.83]**	**80.30 [77.99, 82.66]**	**85.07 [82.81, 87.12]**	**84.41 [82.93, 85.81]**	**71.01 [69.40, 72.51]**	**65.68 [60.65, 70.56]**	**92.00 [90.07, 93.65]**	**79.15 [78.18, 80.09]**	–	–
Base: MobileNet	70.75 [68.64, 73.07]	74.07 [72.24, 75.91]	73.60 [71.04, 76.13]	80.25 [77.77, 82.57]	80.16 [78.57, 81.70]	65.89 [64.03, 67.67]	64.64 [59.56, 69.71]	83.57 [80.46, 86.35]	74.12 [73.06, 75.20]	1E−04 / 8E−04	5.00 [4.29, 5.69]
Base: ViT	73.44 [71.22, 75.68]	76.92 [75.23, 78.57]	77.59 [74.96, 79.92]	83.19 [80.55, 85.38]	82.57 [80.98, 84.05]	68.94 [67.30, 70.58]	60.76 [55.21, 65.82]	90.68 [88.67, 92.38]	76.76 [75.74, 77.81]	1E−04 / 8E−04	2.35 [1.84, 2.95]
Ensemble-MLP	74.20 [71.94, 76.42]	77.89 [76.14, 79.48]	78.51 [75.72, 80.88]	84.86 [82.66, 86.91]	83.80 [82.15, 85.28]	69.81 [68.07, 71.41]	59.61 [53.36, 65.86]	90.06 [87.64, 92.03]	77.34 [76.25, 78.44]	1E−02 / 2E−02	1.75 [0.85, 2.61]
Ensemble-Max	74.14 [72.01, 76.31]	77.67 [76.02, 79.35]	78.54 [75.75, 80.90]	84.08 [81.84, 86.18]	83.24 [81.61, 84.69]	69.48 [67.67, 71.13]	63.87 [58.70, 69.03]	89.09 [86.74, 91.19]	77.51 [76.49, 78.57]	1E−04 / 8E−04	1.59 [1.21, 2.00]
Ensemble-MLP (Two Layer)	74.29 [72.08, 76.47]	77.95 [76.25, 79.57]	78.43 [75.72, 80.83]	84.73 [82.61, 86.84]	83.99 [82.40, 85.47]	69.68 [67.96, 71.25]	61.33 [55.62, 67.38]	89.71 [87.22, 91.74]	77.51 [76.45, 78.62]	2E−02 / 2E−02	1.58 [0.72, 2.42]
Ensemble-Soft Gating	75.13 [73.02, 77.32]	78.07 [76.39, 79.70]	79.47 [76.75, 81.76]	85.02 [82.78, 87.05]	83.97 [82.38, 85.39]	70.02 [68.30, 71.64]	66.80 [61.75, 71.89]	90.39 [88.14, 92.27]	78.61 [77.58, 79.64]	3E−04 / 15E−04	0.49 [0.35, 0.65]
DE-GF	75.19 [73.07, 77.38]	78.11 [76.44, 79.72]	79.58 [76.91, 81.82]	85.04 [82.85, 87.10]	83.93 [82.33, 85.35]	70.02 [68.29, 71.65]	66.67 [61.80, 71.67]	90.44 [88.26, 92.31]	78.62 [77.63, 79.63]	8E−04 / 32E−04	0.48 [0.34, 0.63]
Grid-CSF	75.17 [73.07, 77.37]	78.13 [76.46, 79.75]	79.58 [76.92, 81.85]	85.29 [82.88, 87.06]	83.94 [82.33, 85.38]	70.33 [68.31, 71.65]	68.56 [61.56, 72.05]	90.49 [88.31, 92.35]	78.67 [77.67, 79.68]	1E−03 / 32E−04	0.44 [0.31, 0.56]
DE-CSF	**75.60 [73.53, 77.78]**	**78.37 [76.69, 79.97]**	**80.03 [77.46, 82.27]**	**85.49 [83.35, 87.44]**	**84.22 [82.66, 85.63]**	**70.36 [68.63, 72.00]**	**67.92 [62.84, 73.07]**	**90.84 [88.77, 92.64]**	**79.10 [78.13, 80.06]**	–	–
Base: DenseNet	72.70 [70.58, 74.80]	75.19 [73.42, 76.98]	76.93 [74.42, 79.50]	81.20 [78.83, 83.55]	81.47 [79.90, 82.96]	68.77 [67.03, 70.43]	60.75 [56.00, 65.01]	89.80 [87.49, 91.92]	75.85 [74.87, 76.74]	1E−04 / 8E−04	4.51 [3.83, 5.17]
Base: Swin-T	76.29 [74.24, 78.42]	78.15 [76.49, 79.68]	80.51 [77.95, 82.79]	85.78 [83.64, 87.76]	85.34 [83.91, 86.62]	71.47 [69.93, 73.16]	60.53 [55.38, 65.61]	90.92 [88.93, 92.79]	78.62 [77.66, 79.60]	1E−04 / 8E−04	1.74 [1.29, 2.19]
Ensemble-MLP (Two Layer)	77.69 [75.57, 79.80]	78.80 [77.22, 80.35]	81.13 [78.77, 83.40]	85.42 [83.32, 87.52]	85.37 [83.93, 86.65]	72.21 [70.63, 73.80]	54.80 [47.92, 60.81]	91.13 [88.78, 92.99]	78.32 [77.22, 79.34]	4E−03 / 16E−03	2.04 [1.14, 2.91]
Ensemble-MLP	77.99 [75.88, 80.11]	78.80 [77.19, 80.32]	81.25 [78.84, 83.47]	85.42 [83.28, 87.50]	85.34 [83.89, 86.64]	72.29 [70.66, 73.91]	59.10 [52.57, 65.36]	90.66 [88.31, 92.59]	78.86 [77.78, 79.91]	3E−02 / 3E−02	1.51 [0.65, 2.44]
Ensemble-Max	77.04 [74.91, 79.12]	77.87 [76.20, 79.41]	80.98 [78.47, 83.24]	84.70 [82.47, 86.91]	84.63 [83.12, 85.97]	71.52 [69.87, 73.12]	63.00 [57.63, 68.09]	91.43 [89.23, 93.31]	78.90 [77.92, 79.80]	1E−04 / 8E−04	1.47 [1.18, 1.73]
Ensemble-Soft Gating	78.11 [76.17, 80.11]	78.50 [76.93, 80.04]	81.54 [79.20, 83.78]	86.03 [83.85, 88.12]	85.21 [83.75, 86.51]	72.39 [70.74, 73.96]	64.24 [58.86, 69.20]	92.01 [89.98, 93.70]	79.75 [78.82, 80.65]	1E−03 / 5E−03	0.61 [0.38, 0.85]
DE-GF	77.95 [76.05, 80.02]	78.92 [77.37, 80.50]	81.56 [79.21, 83.85]	85.98 [83.81, 88.02]	85.53 [84.07, 86.81]	72.28 [70.68, 73.87]	64.70 [59.65, 69.76]	92.00 [89.94, 93.70]	79.87 [78.92, 80.77]	4E−03 / 16E−03	0.50 [0.34, 0.66]
Grid-CSF	77.98 [76.01, 79.98]	78.89 [77.36, 80.49]	81.68 [79.24, 83.87]	85.14 [83.56, 88.19]	85.67 [84.21, 86.95]	72.30 [70.70, 73.86]	65.37 [60.38, 70.02]	92.61 [90.98, 93.73]	79.92 [79.08, 80.91]	1E−02 / 2E−02	0.35 [0.27, 0.44]
DE-CSF	**78.26 [76.38, 80.30]**	**79.22 [77.69, 80.76]**	**82.09 [79.65, 84.24]**	**86.48 [84.39, 88.42]**	**85.87 [84.43, 87.14]**	**72.59 [71.02, 74.16]**	**65.86 [60.89, 70.80]**	**92.52 [90.68, 94.08]**	**80.36 [79.46, 81.25]**	–	–
Base: MobileNet	70.75 [68.64, 73.07]	74.07 [72.24, 75.91]	73.60 [71.04, 76.13]	80.25 [77.77, 82.57]	80.16 [78.57, 81.70]	65.89 [64.03, 67.67]	64.64 [59.56, 69.71]	83.57 [80.46, 86.35]	74.12 [73.06, 75.20]	1E−04 / 8E−04	6.39 [5.77, 7.08]
Base: Swin-T	76.33 [74.20, 78.41]	78.13 [76.50, 79.68]	80.48 [77.95, 82.78]	85.77 [83.62, 87.75]	85.34 [83.92, 86.66]	71.49 [69.94, 73.12]	60.48 [55.42, 65.67]	90.91 [88.95, 92.79]	78.62 [77.67, 79.60]	1E−04 / 8E−04	1.87 [1.23, 2.53]
Ensemble-MLP (Two Layer)	77.18 [75.06, 79.35]	78.72 [77.13, 80.39]	80.97 [78.51, 83.18]	85.61 [83.38, 87.62]	85.37 [83.92, 86.71]	71.13 [69.44, 72.71]	58.59 [52.61, 64.58]	90.08 [87.77, 92.06]	78.46 [77.40, 79.48]	3E−03 / 12E−03	2.04 [0.94, 3.08]
Ensemble-MLP	76.77 [74.65, 78.90]	78.53 [76.87, 80.22]	81.50 [79.19, 83.80]	85.10 [82.89, 87.27]	85.62 [84.17, 86.95]	71.22 [69.44, 72.82]	58.51 [51.86, 65.08]	90.53 [88.17, 92.47]	78.47 [77.35, 79.55]	4E−03 / 12E−03	2.01 [0.86, 3.14]
Ensemble-Max	75.70 [73.55, 77.94]	77.83 [76.16, 79.43]	80.69 [78.02, 82.90]	85.45 [83.28, 87.43]	84.58 [83.05, 85.97]	70.58 [68.79, 72.21]	64.75 [59.59, 69.95]	90.50 [88.32, 92.44]	78.76 [77.78, 79.74]	1E−04 / 8E−04	1.74 [1.32, 2.19]
Ensemble-Soft Gating	77.17 [75.10, 79.26]	78.51 [76.90, 80.10]	82.12 [79.64, 84.24]	86.40 [84.22, 88.32]	85.69 [84.29, 87.02]	71.92 [70.27, 73.50]	64.77 [59.70, 69.99]	91.43 [89.48, 93.22]	79.75 [78.76, 80.72]	5E−04 / 25E−04	0.74 [0.37, 1.13]
DE-GF	77.14 [75.11, 79.23]	78.66 [77.03, 80.23]	82.17 [79.68, 84.35]	86.46 [84.32, 88.35]	85.76 [84.35, 87.07]	71.91 [70.24, 73.46]	65.26 [60.37, 70.45]	91.46 [89.52, 93.26]	79.85 [78.87, 80.82]	3E−03 / 12E−03	0.65 [0.28, 1.00]
Grid-CSF	77.16 [75.12, 79.21]	78.65 [77.00, 80.26]	82.17 [79.69, 84.37]	86.45 [84.30, 88.36]	85.76 [84.35, 87.07]	71.90 [70.23, 73.46]	67.50 [62.41, 72.64]	91.46 [89.51, 93.26]	80.13 [79.16, 81.10]	3E−02 / 3E−02	0.37 [0.24, 0.50]
DE-CSF	**77.59 [75.59, 79.66]**	**78.97 [77.35, 80.57]**	**82.65 [80.17, 84.78]**	**86.68 [84.55, 88.54]**	**85.96 [84.56, 87.27]**	**72.21 [70.53, 73.75]**	**68.13 [62.93, 73.08]**	**91.83 [89.97, 93.58]**	**80.50 [79.53, 81.42]**	–	–

Non-Equivalent Reference Results from Literature*											
Model	**Nodule**	**Atelectasis**	**Mass**	**Pneumothorax**	**Effusion**	**Infiltration**	**Pneumonia**	**Cardiomegaly**	**Mean**	**p-value / Holm**	**ΔAUC**
ResNet50 [17]	68.94	73.87	77.04	83.21	81.79	68.85	59.21	89.18	75.26	-	-
SEResNet50 [17]	69.34	73.81	75.92	85.02	81.40	69.78	60.65	89.72	75.71	-	-
Conformer [17]	70.66	72.68	76.14	85.96	80.62	69.73	63.07	90.69	76.19	-	-
ConvNeXt [17]	72.73	73.31	77.36	85.34	82.77	68.87	65.84	**91.42**	77.21	-	-
RepVGG [17]	71.17	73.57	77.27	81.21	80.36	70.25	63.04	89.56	75.80	-	-
DeiT [17]	70.66	70.19	74.78	80.91	77.17	67.16	58.34	80.81	72.50	-	-
CTransCNN [17]	74.17	74.84	78.06	87.25	83.52	70.65	62.99	89.25	77.59	-	-
DualCheXN [40]	**79.60**	78.40	83.80	87.60	83.10	70.50	72.70	88.80	80.60	-	-
ImageGCN [41]	76.80	80.20	84.30	90.00	87.40	70.20	71.50	89.40	81.22	-	-
CheX-DS [13]	76.80	**81.80**	**85.60**	**90.70**	**88.80**	**72.10**	**78.30**	90.10	**83.00**	-	-

DenseNet: CNN-based baseline model. **MobileNet**: CNN-based baseline model. **VIT**: Transformer-based baseline model. **Swin-T**: Swin Transformer as hybrid backbone. **Ensemble-Max**: Heuristic ensemble that combines CNN and Transformer predictions using a maximum operation. **Ensemble-MLP**: Decision-level CNN–Transformer ensemble with a single-layer MLP-based late fusion module. **Ensemble-MLP (Two Layer)**: Decision-level CNN–Transformer ensemble with a two-layer MLP-based late fusion module. **Ensemble-Soft Gating**: Decision-level CNN–Transformer ensemble in which a gating network learns soft, sample-dependent weights to adaptively combine the probability vectors produced by the base models. **Grid-CSF**: Grid Search–Based Class-Specific Fusion. CNN–Transformer ensemble with class-specific fusion weights optimized using a grid search algorithm. **DE-GF**: Global Fusion. CNN–Transformer ensemble optimized using Differential Evolution with globally shared fusion weights. **DE-CSF**: Class-Specific Fusion. CNN–Transformer ensemble optimized using Differential Evolution with class-specific fusion weights, where independent weights are learned for each disease class. All performance metrics are reported with 95% confidence intervals obtained via bootstrapping (1000 iterations).

*The methods listed in the final block were not reproduced under our experimental protocol and are included only as non-equivalent reference results.

Statistical significance ($p < 0.05$) was assessed using a permutation test with 10000 iterations while preserving the integrity of each patient's eight-label prediction vector. The resulting p-values were adjusted using the Holm method within each backbone-combination block.

gains are consistently observed across most disease categories and backbone configurations. The statistical analysis suggests that the observed improvements are unlikely to result from random variation and instead indicate a consistent performance gain of the proposed method. The p-values obtained from the paired permutation test do not necessarily follow the ranking of the methods based on macro-average AUC values. This is expected because the paired permutation test assesses the statistical significance of differences in macro-average AUC while preserving the patient-level pairing between model predictions. Consequently, statistical significance depends not only on the magnitude of the observed performance difference but also on the variability of the paired predictions across the test patients.

In Fig. 1, we compare the ROC curves of the fusion strategies under two backbone configurations: (i) DenseNet + ViT and (ii) DenseNet + Swin-T. For each configuration, ROC curves are plotted for the individual backbones as well as their fusion variants (Ensemble-Max, Ensemble-MLP with one- and two-layer architectures, Ensemble-Soft Gating, DE-GF, and DE-CSF). The macro-average AUC, defined as the mean of class-wise AUC values, is also reported to summarize overall performance. In the DenseNet + ViT setting, fusion strategies consistently outperform individual backbones. Among these, the DE-based approaches exhibit higher ROC curves, indicating improved discriminative capability across disease classes. A similar outcome is observed in the DenseNet + Swin-T configuration. While the Swin-T backbone already provides stronger performance, the DE-based fusion strategies further refine the decision boundaries, leading to improved performance across disease classes.

Among the evaluated fusion methods, the DE-CSF strat-

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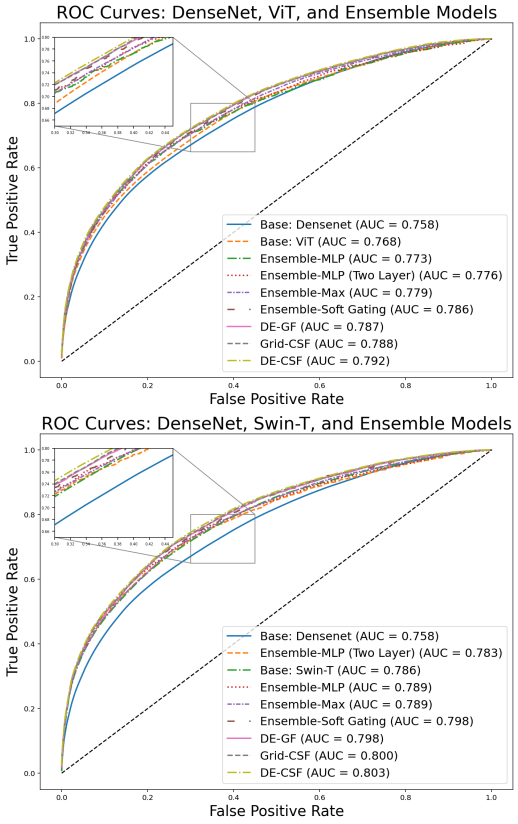


FIGURE 1. **ROC curves of fusion strategies under two backbone configurations: (up) DenseNet + ViT and (bottom) DenseNet + Swin-T. For each configuration, individual backbones and their fusion variants (Ensemble-Max, Ensemble-MLP, Grid-CSF, DE-GF and DE-CSF) are compared.**

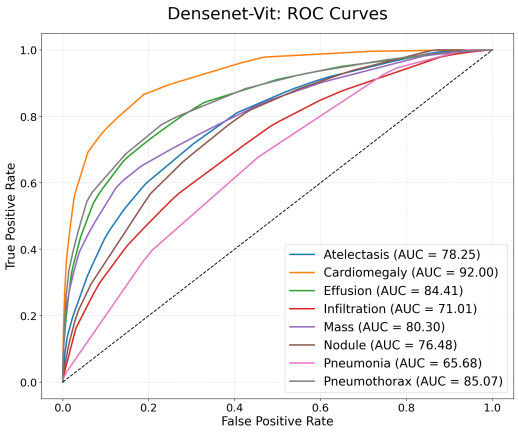


FIGURE 2. **Class-wise ROC curves of the DE-CSF (DenseNet + ViT) fusion strategy for each thoracic disease category.**

egy achieves the highest mean AUC across all backbone configurations. Therefore, we present the class-wise ROC curves of the DE-CSF model for each thoracic pathology in Figure 2. The ROC curves demonstrate that the DE-CSF framework achieves particularly high AUC values for cardiomegaly, pleural effusion, and pneumothorax. In contrast, performance decreases for visually subtle diseases such as pneumonia and infiltration. These results are consistent with the results reported in Table 9, where most models exhibit comparatively lower AUC values for these categories.

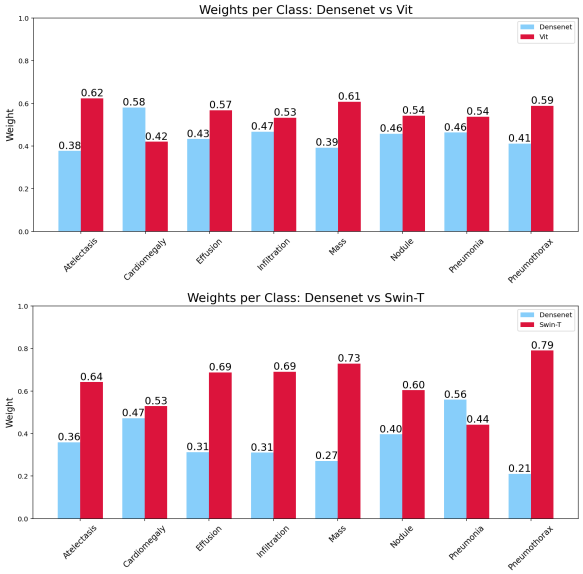


FIGURE 3. **Class-specific fusion weights learned by the DE-CSF strategy under two backbone configurations: (top) DenseNet + ViT and (bottom) DenseNet + SwinT. The learned weights illustrate the adaptive, disease-specific contribution of CNN and Transformer architectures across different model pairings.**

Figure 3 presents the class-specific fusion weights learned by the DE-CSF framework for the DenseNet+ViT and DenseNet+Swin-T backbone configurations. The learned weights demonstrate that the relative contributions of CNN- and Transformer-based architectures vary considerably across disease categories and backbone combinations. In the DenseNet+ViT configuration, the CNN backbone receives higher weights for several disease classes, including Cardiomegaly, Infiltration, Mass, Nodule, and Pneumonia, whereas the DenseNet+Swin-T configuration generally assigns higher weights to the Swin-T backbone for most disease categories. These observations suggest that the discriminative advantage of each architecture varies across disease classes and is influenced by disease characteristics, dataset properties, backbone representational capacity, and optimization dynamics. Since each backbone pair defines a different representational and optimization landscape, the DE-CSF framework adaptively adjusts the class-specific fusion weights. These findings support the use of adaptive and data-driven fusion strategies rather than relying on fixed or manually predefined weighting schemes.

In order to further analyze the performance of the fusion methods, AUPRC was computed to provide a more assessment of precision-recall trade-offs across disease categories. Table 10 presents the class-wise AUPRC performance of the evaluated models. Compared with the AUC results, the differences among the fusion strategies are generally smaller, indicating that improvements in precision-recall performance are more difficult to achieve on this highly imbalanced multi-

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TABLE 10. Comparison of AUPRC performance across different diseases on the NIH CXR dataset.

Results Obtained Under Our Experimental Protocol

Model	Nodule	Atelectasis	Mass	Pneumothorax	Effusion	Infiltration	Pneumonia	Cardiomegaly	Mean	p-value / Holm	ΔAUPRC
Base: DenseNet	0.288 [0.260, 0.320]	0.498 [0.460, 0.530]	0.399 [0.360, 0.450]	0.424 [0.370, 0.480]	0.658 [0.630, 0.690]	0.611 [0.590, 0.640]	0.033 [0.030, 0.040]	0.506 [0.430, 0.570]	0.427 [0.414, 0.441]	1E−04 / 8E−04	0.059 [0.051, 0.067]
Base: ViT	0.310 [0.280, 0.340]	0.523 [0.490, 0.560]	0.399 [0.350, 0.450]	0.476 [0.420, 0.520]	0.677 [0.650, 0.700]	0.622 [0.600, 0.650]	0.034 [0.030, 0.050]	0.478 [0.410, 0.550]	0.440 [0.426, 0.455]	1E−04 / 8E−04	0.046 [0.040, 0.052]
Ensemble-Max	0.334 [0.300, 0.370]	0.533 [0.500, 0.560]	0.442 [0.400, 0.490]	0.492 [0.440, 0.540]	0.700 [0.670, 0.730]	0.634 [0.610, 0.660]	0.037 [0.030, 0.060]	0.534 [0.460, 0.600]	0.463 [0.450, 0.478]	1E−04 / 8E−04	0.023 [0.019, 0.027]
Ensemble-MLP	0.329 [0.290, 0.370]	0.550 [0.520, 0.580]	0.465 [0.420, 0.510]	0.530 [0.480, 0.580]	0.715 [0.690, 0.740]	**0.650 [0.630, 0.680]**	0.034 [0.030, 0.050]	0.561 [0.490, 0.620]	0.479 [0.466, 0.494]	7E−01 / 1E+00	0.007 [0.004, 0.010]
Ensemble-MLP (Two Layer)	0.348 [0.310, 0.390]	0.551 [0.520, 0.580]	0.480 [0.430, 0.530]	0.530 [0.480, 0.580]	**0.718 [0.690, 0.740]**	**0.649 [0.620, 0.670]**	0.033 [0.020, 0.050]	**0.570 [0.500, 0.630]**	**0.485 [0.471, 0.500]**	9E−01 / 1E+00	0.001 [−0.002, 0.005]
Ensemble-Soft Gating	0.348 [0.310, 0.390]	0.553 [0.520, 0.590]	**0.484 [0.440, 0.530]**	0.529 [0.480, 0.580]	0.710 [0.680, 0.740]	0.645 [0.620, 0.670]	**0.041 [0.030, 0.070]**	**0.571 [0.500, 0.630]**	**0.485 [0.472, 0.500]**	1E+00 / 1E+00	0.001 [−0.002, 0.003]
DE-GF	**0.350 [0.310, 0.390]**	**0.555 [0.520, 0.590]**	**0.483 [0.440, 0.530]**	**0.533 [0.480, 0.590]**	0.712 [0.680, 0.740]	0.645 [0.620, 0.670]	**0.040 [0.030, 0.070]**	0.563 [0.490, 0.630]	**0.485 [0.472, 0.500]**	1E+00 / 1E+00	0.001 [0.000, 0.002]
Grid-CSF	**0.350 [0.310, 0.390]**	0.558 [0.520, 0.590]	**0.484 [0.440, 0.530]**	**0.535 [0.480, 0.590]**	0.712 [0.680, 0.740]	0.646 [0.620, 0.670]	0.040 [0.030, 0.070]	0.565 [0.490, 0.630]	**0.486 [0.473, 0.501]**	1E+00 / 1E+00	0.001 [−0.002, 0.003]
DE-CSF	**0.350 [0.310, 0.390]**	**0.556 [0.520, 0.590]**	**0.484 [0.440, 0.530]**	**0.534 [0.480, 0.590]**	0.712 [0.680, 0.740]	0.645 [0.620, 0.670]	**0.040 [0.030, 0.070]**	0.565 [0.490, 0.630]	**0.486 [0.473, 0.501]**	–	–
Base: MobileNet	0.250 [0.230, 0.280]	0.488 [0.450, 0.520]	0.332 [0.290, 0.380]	0.425 [0.370, 0.470]	0.619 [0.590, 0.650]	0.583 [0.560, 0.610]	0.044 [0.030, 0.070]	0.381 [0.320, 0.440]	0.390 [0.377, 0.405]	1E−04 / 8E−04	0.091 [0.081, 0.101]
Base: ViT	0.310 [0.280, 0.340]	0.523 [0.490, 0.560]	0.399 [0.350, 0.450]	0.476 [0.420, 0.520]	0.677 [0.650, 0.700]	0.622 [0.600, 0.650]	0.034 [0.030, 0.050]	0.478 [0.410, 0.550]	0.440 [0.426, 0.455]	1E−04 / 8E−04	0.042 [0.036, 0.049]
Ensemble-Max	0.311 [0.280, 0.350]	0.541 [0.510, 0.570]	0.425 [0.380, 0.470]	0.503 [0.450, 0.550]	0.687 [0.660, 0.710]	0.628 [0.600, 0.650]	0.046 [0.030, 0.080]	0.494 [0.430, 0.560]	0.454 [0.441, 0.469]	1E−04 / 8E−04	0.027 [0.023, 0.032]
Ensemble-MLP	0.326 [0.290, 0.370]	0.559 [0.530, 0.590]	0.448 [0.400, 0.500]	**0.540 [0.490, 0.590]**	0.709 [0.680, 0.740]	**0.639 [0.610, 0.660]**	0.047 [0.030, 0.080]	0.544 [0.470, 0.610]	0.477 [0.462, 0.492]	8E−04 / 1E−00	0.005 [0.000, 0.010]
Ensemble-MLP (Two Layer)	0.330 [0.290, 0.370]	0.559 [0.530, 0.590]	0.447 [0.400, 0.500]	**0.539 [0.480, 0.590]**	**0.711 [0.680, 0.740]**	**0.639 [0.610, 0.660]**	0.044 [0.030, 0.080]	0.541 [0.470, 0.610]	0.476 [0.462, 0.492]	7E−04 / 1E−00	0.005 [0.001, 0.010]
Ensemble-Soft Gating	**0.341 [0.300, 0.380]**	**0.564 [0.530, 0.600]**	**0.467 [0.420, 0.520]**	**0.540 [0.490, 0.590]**	**0.711 [0.680, 0.740]**	**0.640 [0.610, 0.670]**	**0.060 [0.040, 0.110]**	**0.551 [0.480, 0.620]**	**0.484 [0.470, 0.500]**	1E−00 / 1E−00	-0.003 [-0.005, -0.002]
DE-GF	0.337 [0.300, 0.380]	0.562 [0.530, 0.590]	0.462 [0.410, 0.510]	**0.539 [0.480, 0.590]**	0.709 [0.680, 0.740]	**0.639 [0.610, 0.670]**	0.056 [0.040, 0.100]	0.543 [0.470, 0.610]	0.481 [0.467, 0.496]	1E−00 / 1E−00	0.000 [0.000, 0.001]
Grid-CSF	0.339 [0.300, 0.380]	0.562 [0.530, 0.590]	0.462 [0.410, 0.510]	0.539 [0.480, 0.590]	0.709 [0.680, 0.740]	0.639 [0.610, 0.670]	0.057 [0.040, 0.100]	0.544 [0.480, 0.610]	0.481 [0.467, 0.497]	1E−00 / 1E−00	0.000 [0.000, 0.000]
DE-CSF	0.338 [0.300, 0.380]	0.562 [0.530, 0.590]	0.461 [0.410, 0.510]	**0.539 [0.480, 0.590]**	0.708 [0.680, 0.740]	**0.639 [0.610, 0.670]**	0.057 [0.040, 0.100]	0.544 [0.480, 0.610]	0.481 [0.467, 0.497]	–	–
Base: DenseNet	0.288 [0.260, 0.320]	0.498 [0.460, 0.530]	0.399 [0.360, 0.450]	0.424 [0.370, 0.480]	0.658 [0.630, 0.690]	0.611 [0.590, 0.640]	0.033 [0.030, 0.040]	0.506 [0.430, 0.570]	0.427 [0.414, 0.441]	1E−04 / 8E−04	0.082 [0.073, 0.09]
Base: Swin-T	0.367 [0.330, 0.410]	0.534 [0.500, 0.570]	0.491 [0.440, 0.540]	0.533 [0.480, 0.580]	0.697 [0.670, 0.720]	0.646 [0.620, 0.670]	0.033 [0.030, 0.050]	0.516 [0.450, 0.590]	0.477 [0.464, 0.491]	1E−04 / 8E−04	0.032 [0.027, 0.037]
Ensemble-Max	0.382 [0.340, 0.420]	0.542 [0.510, 0.570]	0.511 [0.460, 0.560]	0.532 [0.480, 0.580]	0.708 [0.680, 0.730]	0.651 [0.630, 0.670]	0.036 [0.030, 0.050]	0.559 [0.490, 0.630]	0.490 [0.478, 0.504]	2E−04 / 12E−04	0.018 [0.014, 0.023]
Ensemble-MLP	0.397 [0.360, 0.440]	**0.563 [0.530, 0.600]**	**0.531 [0.480, 0.580]**	0.568 [0.520, 0.620]	**0.724 [0.700, 0.750]**	0.665 [0.640, 0.690]	0.034 [0.030, 0.050]	0.581 [0.510, 0.650]	0.508 [0.495, 0.522]	1E−00 / 1E−00	0.001 [−0.002, 0.004]
Ensemble-MLP (Two Layer)	0.393 [0.350, 0.430]	**0.563 [0.530, 0.600]**	0.527 [0.480, 0.570]	0.566 [0.510, 0.620]	**0.723 [0.690, 0.750]**	**0.666 [0.640, 0.690]**	0.028 [0.020, 0.040]	**0.586 [0.520, 0.650]**	0.507 [0.493, 0.521]	9E−01 / 1E−00	0.002 [0.000, 0.005]
Ensemble-Soft Gating	**0.399 [0.360, 0.440]**	0.561 [0.530, 0.590]	0.529 [0.480, 0.580]	**0.578 [0.530, 0.630]**	0.721 [0.690, 0.750]	0.665 [0.640, 0.690]	**0.041 [0.030, 0.060]**	0.576 [0.510, 0.640]	**0.509 [0.496, 0.523]**	1E−00 / 1E−00	0.000 [−0.003, 0.002]
Grid-CSF	0.392 [0.350, 0.430]	**0.562 [0.530, 0.590]**	0.528 [0.480, 0.570]	0.570 [0.520, 0.620]	0.722 [0.690, 0.750]	0.663 [0.640, 0.690]	0.039 [0.030, 0.060]	0.584 [0.520, 0.650]	0.507 [0.494, 0.522]	1E−00 / 1E−00	0.000 [-0.001, 0.001]
DE-GF	0.392 [0.350, 0.430]	**0.562 [0.530, 0.590]**	0.528 [0.480, 0.570]	0.570 [0.520, 0.620]	0.722 [0.690, 0.750]	0.663 [0.640, 0.690]	0.039 [0.030, 0.060]	0.584 [0.520, 0.650]	0.507 [0.494, 0.522]	1E−00 / 1E−00	0.001 [0.000, 0.003]
DE-CSF	0.392 [0.350, 0.430]	**0.563 [0.530, 0.590]**	**0.530 [0.480, 0.580]**	0.572 [0.520, 0.620]	**0.723 [0.690, 0.750]**	0.663 [0.640, 0.690]	**0.040 [0.030, 0.060]**	0.584 [0.520, 0.650]	**0.509 [0.496, 0.523]**	–	–
Base: MobileNet	0.250 [0.230, 0.280]	0.488 [0.450, 0.520]	0.332 [0.290, 0.380]	0.425 [0.370, 0.470]	0.619 [0.590, 0.650]	0.583 [0.560, 0.610]	0.044 [0.030, 0.070]	0.381 [0.320, 0.440]	0.390 [0.377, 0.405]	1E+04 / 8E+04	0.115 [0.104, 0.125]
Base: Swin-T	0.367 [0.330, 0.410]	0.534 [0.500, 0.570]	0.491 [0.440, 0.540]	0.533 [0.480, 0.580]	0.697 [0.670, 0.720]	0.646 [0.620, 0.670]	0.033 [0.030, 0.050]	0.516 [0.450, 0.590]	0.477 [0.464, 0.491]	1E+04 / 8E+04	0.029 [0.024, 0.034]
Ensemble-Max	0.362 [0.320, 0.400]	0.544 [0.510, 0.580]	0.500 [0.450, 0.550]	0.552 [0.500, 0.600]	0.696 [0.670, 0.720]	0.642 [0.620, 0.670]	0.044 [0.030, 0.070]	0.522 [0.450, 0.590]	0.483 [0.469, 0.497]	1E+04 / 8E+04	0.023 [0.018, 0.027]
Ensemble-MLP	0.376 [0.340, 0.420]	**0.567 [0.540, 0.600]**	0.531 [0.480, 0.580]	0.559 [0.510, 0.610]	**0.721 [0.690, 0.750]**	0.657 [0.630, 0.680]	0.038 [0.030, 0.060]	0.550 [0.480, 0.620]	0.500 [0.487, 0.514]	8E+01 / 1E+00	0.005 [0.001, 0.010]
Ensemble-MLP (Two Layer)	0.372 [0.330, 0.420]	**0.568 [0.540, 0.600]**	0.525 [0.480, 0.570]	0.556 [0.500, 0.610]	0.719 [0.690, 0.750]	0.656 [0.630, 0.680]	0.039 [0.030, 0.070]	0.543 [0.480, 0.610]	0.497 [0.484, 0.512]	6E+01 / 1E+00	0.008 [0.004, 0.012]
Ensemble-Soft Gating	**0.386 [0.350, 0.430]**	0.563 [0.530, 0.600]	**0.535 [0.480, 0.580]**	**0.572 [0.520, 0.620]**	0.720 [0.690, 0.750]	**0.660 [0.640, 0.690]**	0.053 [0.040, 0.100]	**0.565 [0.500, 0.630]**	**0.507 [0.493, 0.522]**	1E+00 / 1E+00	-0.002 [-0.004, 0.000]
DE-GF	0.381 [0.340, 0.420]	0.563 [0.530, 0.600]	0.532 [0.480, 0.580]	0.567 [0.510, 0.620]	0.719 [0.690, 0.750]	**0.659 [0.640, 0.680]**	0.050 [0.030, 0.090]	0.556 [0.490, 0.630]	0.504 [0.490, 0.519]	1E+00 / 1E+00	0.001 [0.000, 0.003]
Grid-CSF	0.382 [0.340, 0.420]	0.564 [0.530, 0.600]	0.532 [0.480, 0.580]	0.569 [0.520, 0.620]	0.719 [0.690, 0.750]	0.659 [0.640, 0.680]	**0.056 [0.040, 0.100]**	0.556 [0.490, 0.630]	0.505 [0.491, 0.520]	1E+00 / 1E+00	0.000 [0.000, 0.000]
DE-CSF	0.382 [0.340, 0.420]	0.564 [0.530, 0.600]	0.532 [0.480, 0.580]	0.569 [0.510, 0.620]	0.719 [0.690, 0.750]	**0.659 [0.640, 0.680]**	**0.056 [0.040, 0.100]**	0.556 [0.490, 0.620]	0.505 [0.491, 0.520]	–	–

Non-Equivalent Reference Results from Literature*

Model	Nodule	Atelectasis	Mass	Pneumothorax	Effusion	Infiltration	Pneumonia	Cardiomegaly	Mean	p-value / Holm	ΔAUPRC
*A-GCN-PPS	0.172	0.275	0.237	0.303	0.479	0.347	**0.037**	0.256	0.208	-	-
*A-GCN-APS	0.165	0.243	0.224	0.284	0.384	0.301	0.028	0.212	0.182	-	-
*AlexNet	0.163	0.280	0.229	0.300	0.468	0.335	0.032	0.251	0.197	-	-
*R-GCN-PPS	0.186	0.281	0.238	0.344	0.478	0.338	**0.038**	0.250	0.222	-	-
*R-GCN-APS	0.175	0.228	0.198	0.277	0.378	0.309	0.025	0.197	0.185	-	-
*ResNet50	0.163	0.288	0.237	0.311	0f.476	0.330	0.033	0.243	0.212	-	-
*V-GCN-PPS	**0.225**	**0.313**	**0.328**	**0.351**	**0.511**	0.340	0.037	**0.267**	**0.255**	-	-
*V-GCN-APS	0.212	0.272	0.291	0.313	0.414	0.317	0.035	0.216	0.220	-	-
*VGGNet16BN	0.206	0.297	0.262	0.328	0.497	0.332	0.037	0.248	0.223	-	-

All performance metrics are reported with 95% confidence intervals obtained via bootstrapping (1000 iterations).
* The results were taken from study [41].

label dataset. Ensemble Soft Gating, Grid-CSF, DE-GF, and DE-CSF consistently rank among the best-performing fusion approaches across all backbone combinations. While Ensemble Soft Gating achieves comparable or, in some backbone combinations, slightly higher mean AUPRC values than the proposed method, Grid-CSF and DE-CSF consistently achieve performance comparable to or slightly better than DE-GF. Moreover, DE-CSF matches or marginally improves upon Grid-CSF across the evaluated backbone combinations.

Although the proposed DE-based fusion methods achieve the best overall AUC performance across different backbone combinations, they do not always obtain the highest AUPRC. This behavior is expected because the DE algorithm optimizes the fusion weights by maximizing the macro-average AUC rather than AUPRC. Nevertheless, DE-GF and DE-CSF consistently achieve competitive AUPRC performance across all backbone combinations, despite not explicitly optimizing this metric. One possible explanation for the slightly higher AUPRC achieved by Soft Gating for some backbone combinations is that Soft Gating employs sample-specific fusion weights, whereas DE-CSF uses fixed class-specific weights. This additional flexibility may better accommodate variations among individual samples and may result in slightly higher AUPRC values for some backbone combinations.

2) Computational Cost Analysis

Table 11 summarizes the computational cost of the evaluated ensemble methods. For learning-based ensemble approaches (Ensemble-MLP, Ensemble-MLP (Two Layer), and Ensemble-Soft Gating), the table reports the training time required to optimize the fusion model. As expected, DE-GF requires the lowest computational cost because it optimizes only one global fusion parameter, DE-CSF requires additional optimization time due to learning independent fusion weights for each disease class. Nevertheless, both DE-based methods operate as lightweight post-processing frameworks without retraining the underlying CNN or Transformer models.

TABLE 11. Computational cost analysis of ensemble methods

Method	Training Requirement	Trainable/Optimization Parameters	Time (ms)
Ensemble-MLP	Yes	808	3744.94 ± 5590.74
Ensemble-MLP (2-Layer)	Yes	1208	5460.62 ± 6796.75
Ensemble-Soft Gating	Yes	610	8307.44 ± 8941.33
DE-GF	No	1	271.47 ± 3.91
Grid-CSF	No	8	2287.39 ± 18.25
DE-CSF	No	8	4164.47 ± 39.00

3) Probability Calibration Analysis

To evaluate the effectiveness of probability calibration prior to late fusion, we compared the Brier scores of all backbone models before and after applying Isotonic Regression. Lower

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Brier scores indicate better calibrated probability estimates. As shown in Table 12, calibration consistently reduced the Brier score for all backbone models and all disease categories. These results indicate that Isotonic Regression produced more reliable probability estimates, providing a stronger foundation for the subsequent late fusion methods.

To understand the contribution of probability calibration from the DE-CSF framework, we performed an ablation study. DE-CSF was evaluated using both the original uncalibrated probability outputs and the isotonic-calibrated probabilities while keeping all remaining components of the framework unchanged. Table 13 presents the class-wise and mean AUC results for both settings. The results indicate that isotonic regression provides more reliable probability estimates and leads to a modest improvement in classification performance.

4) Ablation Study of U-Net-Based Lung Segmentation

To evaluate the contribution of the U-Net-based lung segmentation stage, we conducted an additional ablation study by training and evaluating all baseline models using both the original CXR images and the lung-masked images produced by the U-Net segmentation model. All remaining preprocessing steps, training procedures, and hyperparameter settings were kept identical to ensure a fair comparison. Table 14 summarizes the mean AUC values obtained under the two configurations. Across all evaluated backbone architectures, lung masking consistently improves classification performance. These results indicate that removing irrelevant anatomical structures outside the lung region enables the models to focus on pulmonary information, resulting in more discriminative feature representations.

VI. DISCUSSION AND CONCLUSIONS

This study presented DE-based optimization frameworks for decision-level fusion of CNN and Transformer models for thoracic disease classification from CXR images. Motivated by the complementary strengths of these architectures, we investigate whether optimization-driven fusion can provide more robust disease-wise performance. We evaluated two DE-based fusion formulations: a globally shared fusion strategy (DE-GF) and a class-specific fusion strategy (DE-CSF). Unlike conventional ensemble methods with fixed or heuristic weighting schemes, the proposed approach learns fusion coefficients through evolutionary optimization, enabling adaptive integration between CNN and Transformer models.

The experiments demonstrated that DE-based optimization consistently improved classification performance over individual CNN and Transformer backbones and compared decision-level fusion methods. Among the evaluated approaches, DE-CSF consistently achieved the highest mean AUC across all backbone combinations, outperforming both globally shared fusion (DE-GF) and grid search-based class-specific fusion (Grid-CSF), while maintaining comparable AUPRC values and balanced performance across disease categories. Although the improvements over DE-GF and Grid-CSF were generally modest, their consistency across different backbone combinations suggests that Differential Evolution provides a more effective optimization strategy for learning class-specific fusion weights than globally shared weighting or grid search. The learned fusion weights varied across disease categories and backbone combinations, suggesting that the optimal relative contribution of the CNN and Transformer models depends on the predictive behavior of the underlying models. By optimizing independent fusion weights for each disease category, DE-CSF adapts these contributions accordingly, resulting in more balanced classification performance across heterogeneous disease categories. Probability calibration using Isotonic Regression further contributed to reliable decision-level fusion by providing better-calibrated confidence estimates for the optimization process. The proposed optimization introduces only a lightweight post-processing step without requiring retraining of the backbone models. These findings suggest that adaptive, data-driven fusion provides a consistent refinement over globally shared weighting schemes.

To evaluate the effectiveness of the DE-based fusion strategies, we compared them with several ensemble learning methods. Overall, both DE-GF and DE-CSF achieved higher performance compared with the evaluated fusion approaches. Conventional weighted averaging combines model predictions using fixed or globally optimized coefficients, whereas DE-GF automatically determines these coefficients through DE. DE-GF and DE-CSF provide interpretable fusion weights that quantify the contribution of each base model to the final prediction. Compared with constrained ensemble optimization methods, DE provides a flexible derivative-free optimization framework for optimizing continuous fusion weights without requiring gradient information. Unlike stacking-based methods, such as MLP fusion, and mixture-of-experts approaches, such as Soft Gating, the proposed DE-based methods do not require training an additional meta-model or gating network, thereby improving practical applicability. Their smaller number of hyperparameters compared with learnable fusion methods reduces the burden of hyperparameter tuning. Particularly in settings where medical data are limited, the absence of a separate meta-training process improves data efficiency and reduces the risk of overfitting during the fusion stage.

One of the limitations of this study is the lack of external validation, as no suitable independent dataset was available that simultaneously included all disease classes examined in this study. In addition, the NIH CXR dataset has known limitations. Since the image-level labels are automatically extracted from radiology reports using NLP techniques, the dataset is known to contain label noise, particularly false-negative labels resulting from missed positive findings [10]. Previous studies have also reported that models developed on this dataset may exhibit limited generalizability across different clinical environments because the dataset does not fully capture the diversity of patient populations and imaging practices [11]. Nevertheless, since all fusion methods evalu-

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TABLE 12. Class-wise Brier scores of DenseNet, MobileNet, ViT, and Swin-T before and after calibration.

Model	Status	Nodule	Atelectasis	Mass	Pneumothorax	Effusion	Infiltration	Pneumonia	Cardiomegaly	Average
DenseNet	**Uncalibrated**	0.2076	0.2256	0.1817	0.1950	0.2253	0.2599	0.1291	0.1236	**0.1935**
	Calibrated	0.1078	0.1541	0.0843	0.0755	0.1407	0.2188	0.0239	0.0391	**0.1055**
	Δ	-0.0998	-0.0715	-0.0974	-0.1195	-0.0846	-0.0411	-0.1052	-0.0845	**-0.0880**
MobileNet	**Uncalibrated**	0.2194	0.2370	0.1944	0.1952	0.2218	0.2520	0.1232	0.1348	**0.1972**
	Calibrated	0.1104	0.1555	0.0899	0.0753	0.1474	0.2260	0.0238	0.0445	**0.1091**
	Δ	-0.1090	-0.0815	-0.1045	-0.1199	-0.0744	-0.0260	-0.0994	-0.0903	**-0.0881**
ViT	**Uncalibrated**	0.1960	0.2271	0.1628	0.1552	0.1980	0.2528	0.1045	0.1048	**0.1752**
	Calibrated	0.1063	0.1492	0.0829	0.0707	0.1357	0.2182	0.0239	0.0398	**0.1033**
	Δ	-0.0897	-0.0779	-0.0799	-0.0845	-0.0623	-0.0346	-0.0806	-0.0650	**-0.0718**
Swin-T	**Uncalibrated**	0.2072	0.2380	0.6130	0.3920	0.9840	0.2477	0.0510	0.2730	**0.3757**
	Calibrated	0.1014	0.1469	0.0762	0.0658	0.1283	0.2109	0.0239	0.0377	**0.0989**
	Δ	-0.1058	-0.0911	-0.5368	-0.3262	-0.8557	-0.0368	-0.0271	-0.2353	**-0.2768**

Uncalibrated: Calibration performance before applying Isotonic Regression. **Calibrated**: Calibration performance after applying Isotonic Regression. Δ: Difference calculated as Calibrated - Uncalibrated. A negative Δ indicates improved calibration.

TABLE 13. Comparison of DE-CSF performance with and without isotonic probability calibration.

		Results Obtained Under Our Experimental Protocol								
Model Combination	**Calibration**	**Nodule**	**Atelectasis**	**Mass**	**Pneumothorax**	**Effusion**	**Infiltration**	**Pneumonia**	**Cardiomegaly**	**Mean**
DenseNet + ViT	Uncalibrated	74.73 [72.57, 76.83]	77.24 [75.56, 78.85]	78.64 [76.18, 81.17]	84.07 [81.70, 86.29]	83.66 [82.19, 85.07]	69.70 [68.08, 71.36]	59.57 [54.19, 64.56]	90.94 [88.62, 92.84]	77.32 [76.31, 78.30]
DenseNet + ViT	Isotonic	76.48 [74.41, 78.42]	78.25 [76.64, 79.83]	80.30 [77.99, 82.66]	85.07 [82.81, 87.12]	84.41 [82.93, 85.81]	71.01 [69.40, 72.51]	65.68 [60.65, 70.56]	92.00 [90.07, 93.65]	79.15 [78.18, 80.09]
MobileNet + ViT	Uncalibrated	73.88 [71.69, 76.13]	77.25 [75.53, 78.92]	78.23 [75.56, 80.69]	84.08 [81.75, 86.22]	83.28 [81.63, 84.76]	68.96 [67.27, 70.64]	63.29 [57.74, 68.64]	90.07 [87.83, 91.95]	77.38 [76.36, 78.43]
MobileNet + ViT	Isotonic	75.60 [73.53, 77.78]	78.37 [76.69, 79.97]	80.03 [77.46, 82.27]	85.49 [83.35, 87.44]	84.22 [82.66, 85.63]	70.36 [68.63, 72.00]	67.92 [62.84, 73.07]	90.84 [88.77, 92.64]	79.10 [78.13, 80.06]
DenseNet + Swin-T	Uncalibrated	73.88 [71.69, 76.13]	77.25 [75.53, 78.92]	78.23 [75.56, 80.69]	84.08 [81.75, 86.22]	83.28 [81.63, 84.76]	68.96 [67.27, 70.64]	63.29 [57.74, 68.64]	90.07 [87.83, 91.95]	77.38 [76.36, 78.43]
DenseNet + Swin-T	Isotonic	78.26 [76.38, 80.30]	79.22 [77.69, 80.76]	82.09 [79.65, 84.24]	86.48 [84.39, 88.42]	85.87 [84.43, 87.14]	72.59 [71.02, 74.16]	65.86 [60.89, 70.80]	92.52 [90.68, 94.08]	80.36 [79.46, 81.25]
MobileNet + Swin-T	Uncalibrated	75.77 [73.65, 77.98]	78.03 [76.34, 79.60]	81.00 [78.45, 83.37]	85.62 [83.37, 87.66]	85.26 [83.80, 86.57]	71.22 [69.62, 72.88]	62.72 [57.16, 68.22]	90.66 [88.62, 92.50]	78.78 [77.78, 79.77]
MobileNet + Swin-T	Isotonic	77.59 [75.59, 79.66]	78.97 [77.35, 80.57]	82.65 [80.17, 84.78]	86.68 [84.55, 88.54]	85.96 [84.56, 87.27]	72.21 [70.53, 73.75]	68.13 [62.93, 73.08]	91.83 [89.97, 93.58]	80.50 [79.53, 81.42]

All results were obtained using the DE-CSF fusion method. **Uncalibrated** denotes DE-CSF using the raw probability outputs of the base models, whereas **Isotonic** denotes DE-CSF using isotonic-calibrated probability outputs. All AUC values are reported with 95% confidence intervals obtained via bootstrapping with 1000 iterations.

TABLE 14. Ablation study comparing the performance of the baseline models with and without U-Net-based lung segmentation.

Model	Configuration	Mean AUROC (%)	Mean AUPRC
DenseNet	Original	62.80 [61.72, 63.94]	0.190 [0.186, 0.196]
	Lung-masked (U-Net)	75.84 [74.87, 76.74]	0.427 [0.414, 0.441]
MobileNet	Original	71.16 [70.13, 72.22]	0.355 [0.343, 0.369]
	Lung-masked (U-Net)	74.12 [73.06, 75.20]	0.390 [0.377, 0.405]
ViT	Original	55.32 [52.52, 58.25]	0.246 [0.240, 0.256]
	Lung-masked (U-Net)	76.76 [75.74, 77.81]	0.440 [0.426, 0.455]
Swin-T	Original	53.68 [52.62, 54.85]	0.188 [0.185, 0.193]
	Lung-masked (U-Net)	78.62 [77.67, 79.60]	0.477 [0.464, 0.491]

ated in this study were trained and tested on the same dataset under identical experimental conditions, these limitations are expected to affect all methods similarly, thereby preserving the fairness of the comparative evaluation.

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